

Next-Day 24-Hour Stock Status Prediction in Fresh Retail

Architectural Evolution from LSTMs to Hourly Slot-Specific DLinear

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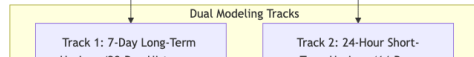
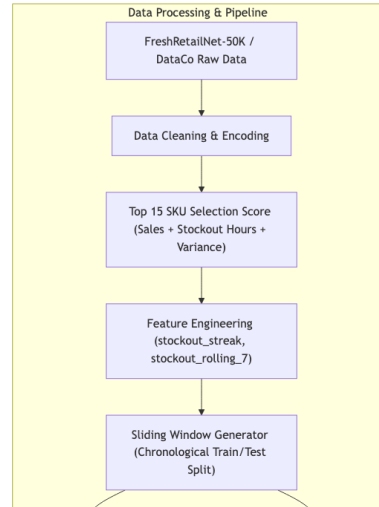
Problem Statement: Next-Day 24-Hour Stock Status Prediction

The Operational Challenge in Fresh Retail

- Perishable fresh goods suffer from **rapid intra-day demand spikes** and supply replenishment delays.
- Stockouts lead to direct financial loss, customer churn, and store operational inefficiency.

Predictive Target Formulation

- Forecast a **24-dimensional binary hourly status**



Literature Review & Related Work

Limitations of Traditional Daily Aggregation

- Traditional retail models forecast 1 scalar daily demand value.
- **Key Flaw:** Hides intraday stockout timing (e.g. stockout at 11 AM vs 8 PM requires vastly different store actions).

Deep Time-Series Transformers

- **PatchTST** (Zeng et al., 2023) & **TFT** (Lim et al., 2021).
- Attention maps point-wise tokens → prone to **overparameterization** (280k+ params) on short sequence splits.

Linear Decomposition Models

- **DLinear** (Zeng et al., 2023): Decomposes sequences into trend (moving-average) and seasonal components.
- Extremely parameter-efficient ($< 15k$ parameters) with strong generalization.

Explainable AI in Time Series

- **Integrated Gradients** (Sundararajan et al., 2017) & **SHAP**.
- Used to inspect feature attributions and guide architectural decoupling.

Data Selection Strategy: FreshRetailNet-50K & SKU Scoring

Unit of Analysis

Time series defined at the individual SKU-store grain:

$$\text{series_id} = \text{city_id} + \text{store_id} + \text{product_id}$$

Why Statistically Filter SKUs?

- Fresh retail datasets contain thousands of dormant SKUs.
- We focus evaluation on **high-impact, stockout-prone SKUs** driving 80%+ of store revenue loss.

SKU Statistical Ranking Formula

$$\begin{aligned}\text{Score} = & 0.40 \cdot \text{Sales} \\ & + 0.25 \cdot \text{StockoutHours} \\ & + 0.20 \cdot \text{StockoutDays} \\ & + 0.15 \cdot \text{SalesStd}\end{aligned}$$

Selected Top 15 SKUs: Represent high-turnover perishable items with high demand variance.

Hourly vs. Daily Forecasting Paradigm

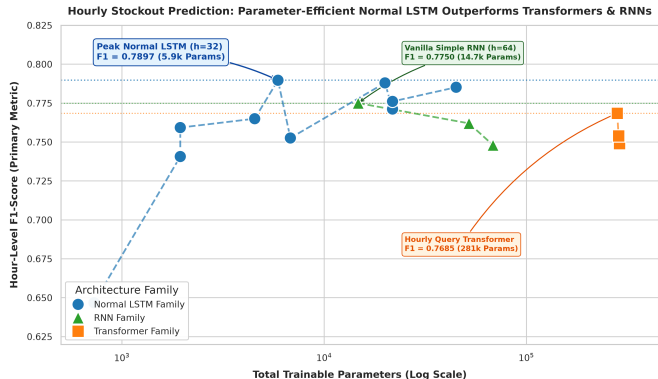
Traditional Daily Aggregation Paradigm

- **Input:** $L = 14$ days of historical daily metrics.
- **Output:** 1 scalar value (Total expected daily sales or binary stockout flag).
- **Loss of Information:** Store managers cannot determine *when* replenishment must occur.

Our 24-Hour Hourly Binary Paradigm

- **Input:** $L = 14$ days of SKU-store daily feature vectors.
- **Output:** 24 binary probabilities $\hat{\mathbf{Y}} \in [0, 1]^{24}$.
- **Actionable Insight:** Identifies peak risk hours (e.g. 11 AM lunch rush & 6 PM evening peak) for proactive restocking.

Model Family Comparison: Simple LSTMs Beat Heavy Transformers



Key Empirical Findings

- **Parameter Efficiency:** Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization:** PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win:** Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability: SHAP & Integrated Gradients Analysis

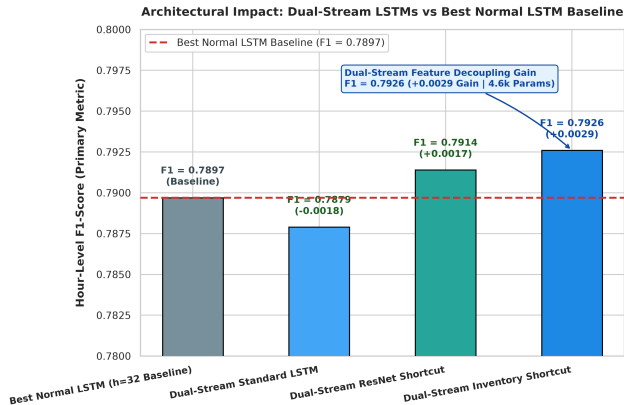
Integrated Gradients (IG) Feature Attribution

- Discovered that **Demand Signals** (sales amount, discounts) and **Inventory Signals** (stockout counts, rolling stockouts) act on different temporal scales.
- Demand features exhibit fast 1–3 day velocity, while inventory features exhibit 7-day weekly seasonality.

Identification of Feature Cross-Talk

- Feeding demand & inventory features into a single shared LSTM recurrent state causes **gradient noise and interference**.
- **Architectural Takeaway:** Inspired the design of **Dual-Stream Decoupling** to isolate sales and inventory signals into parallel streams.

Architectural Innovation 1: Dual-Stream Feature Decoupling



Dual-Stream Mechanism

- Runs **two parallel 16-dim BiLSTMs**:
 - 1 **Demand Stream**: Sales, discounts, sales momentum.
 - 2 **Inventory Stream**: Stockout counts, rolling stockouts.
- Includes additive **ResNet residual shortcuts**.

Empirical Gain

- Boosts F1-Score from **0.7897** (Baseline Normal LSTM) to **0.7926** (Dual-Stream Inventory).

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

$$X_{\text{trend}} = \text{AvgPool}(X)$$

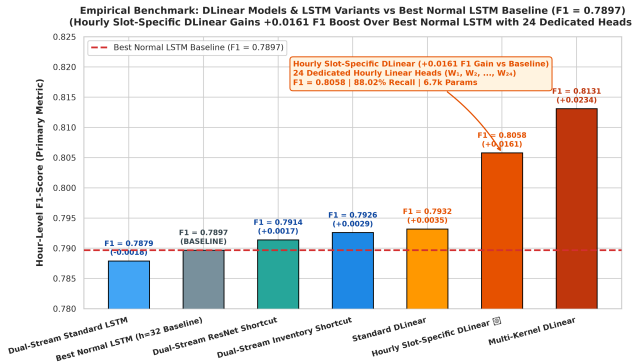
$$X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$Y_{\text{pred}} = W_{\text{trend}}X_{\text{trend}} + W_{\text{seasonal}}X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters:** 6,768 parameters.
- **Validation F1-Score: 0.7932** (Outperforms both Normal LSTMs and heavy Transformers).
- **Stockout Recall: 87.44%.**

Architectural Innovation 2: Hourly Slot-Specific DLinear



24 Dedicated Hourly Linear Heads

Replaces flat matrix multiplication with 24 dedicated hourly linear heads ($W_1, W_2, \dots, W_{24}$):

$$\hat{y}_h = W_{\text{trend},h}X_{\text{trend}} + W_{\text{seasonal},h}X_{\text{seasonal}}$$

Performance Breakthrough

- **Hourly Slot DLinear: 0.8058 F1-Score & 88.02% Recall** (+0.0161 gain over baseline).
- **Multi-Kernel DLinear: 0.8131 F1-Score** ($k = 3, 5, 7$ multi-scale pooling).

Master Empirical Validation Leaderboard

Rank	Architecture Name	Category	Params	F1-Score	Recall	MAE (hrs)
1	Multi-Kernel DLinear	Linear (Multi-Scale)	13,539	0.8131	85.22%	3.86
2	Hourly Slot-Specific DLinear	Linear (24 Heads)	6,744	0.8058	88.02%	4.23
3	Standard DLinear	Linear (Decomp)	6,768	0.7932	87.44%	4.28
4	Dual-Stream Inventory Shortcut	LSTM Family	4,600	0.7926	80.79%	3.96
5	Dual-Stream ResNet Shortcut	LSTM Family	4,728	0.7914	80.94%	4.11
Base	Best Normal LSTM (h=32)	Normal LSTM	5,912	0.7897	81.29%	4.31
6	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%	4.58
7	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%	4.67
8	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%	4.23
9	Hourly Query Transformer	Transformer	281,345	0.7685	82.08%	4.25
10	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%	4.16

Conclusion & Operational Deployment Takeaways

Key Architectural Takeaways

- 1 **Parameter Efficiency Wins:** Compact LSTMs and Linear models (4.6k–13.5k params) outperform 280k+ param Transformers.
- 2 **Feature Decoupling Matters:** Decoupling sales and inventory signals eliminates gradient noise (+0.0029 F1 gain).
- 3 **Dedicated Hourly Heads:** Resolves inter-hour gradient conflict between morning and evening hours (+0.0161 F1 gain).

Deployment Recommendations

- **Primary Production Deployment: Hourly Slot-Specific DLinear** (0.8058 F1, 88.02% Recall, 6.7k params, 0.50 ms latency).
- **High-Accuracy Variant: Multi-Kernel DLinear** (0.8131 F1, 3.86 hrs MAE).
- **Future Scope:** Incorporate real-time intraday point-of-sale stream updates for dynamic threshold adjustment.